RETAIL SALES ANALYTICS DASHBOARD

A Project Report Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

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Declaration

The Project Report entitled "**RETAIL SALES ANALYTICS DASHBOARD**" is a record of Bonafide work of team members Cherukuri Sai Deekshitha – 2320030278,Paladi Raga Amrutha Ratna – 2320030355,Ravuri Bramhani - 2320030104, submitted in partial fulfilment for the award of B. Tech in Computer Engineering to K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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Certificate

This is to certify that the project-based report entitled **"RETAIL SALES ANALYTICS DASHBOARD"** is a Bonafide work carried out and submitted by Cherukuri Sai Deekshitha – 2320030278,Paladi Raga Amrutha Ratna – 2320030355,Ravuri Bramhani - 2320030104, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science Engineering**, at **K L (Deemed to be University)**, during the academic year **2025–2026**.

Signature of the Supervisor

Signature of the HOD **Signature of the External Exami**

ACKNOWLEDGEMENT

The success in this project would not have been possible but for the timely help and guidance rendered by many people. Our wish to express my sincere thanks to all those who has assisted us in one way or the other for the completion of my project.

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We thank all the members of teaching and non-teaching staff members, and who have assisted us directly or indirectly for successful completion of this project. Finally, I sincerely thank my parents, friends and classmates for their kind help and cooperation during my work.

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ABSTRACT

Retail businesses generate large volumes of sales data on a daily basis through transactions, products, and customer interactions. However, traditional systems often face challenges in handling large datasets, providing scalability, and delivering timely insights. This creates the need for a modern cloud-based solution that can efficiently store, process, and analyze retail data for better decision-making.

This project presents a Retail Sales Analytics Dashboard using Microsoft Azure and Power BI. In this system, sales data is stored in Azure Blob Storage, which provides scalable and cost-effective storage for raw data. Azure Data Factory is used to automate the data ingestion process, transferring data from storage into a structured format in Azure Synapse Analytics, where it is processed and organized for analysis.

Finally, Power BI is used to create interactive dashboards that visualize key information such as sales trends, product performance, and regional analysis. The system enables efficient data handling, improved scalability, and clear data visualization, helping retail businesses gain meaningful insights and support data-driven decision-making in a cost-effective and cloud-based environment.

INTRODUCTION

In today's data-driven world, retail organizations continuously generate large volumes of information through their sales operations. Every transaction, whether online or offline, captures important details such as product category, region, sales channel, revenue, and demand patterns. Analyzing this information allows businesses to gain deeper insights into performance, understand customer demand, and make informed strategic decisions.

The **Retail Sales Analytics Dashboard using Azure Cloud** is developed to process and analyze such retail data using a scalable and efficient cloud-based approach. Unlike traditional data analysis methods that require complex infrastructure, this system focuses on simplicity, cost-effectiveness, and ease of use while maintaining high performance.

The system integrates:
• **Azure Data Lake Storage Gen2** → to store large-scale retail datasets securely
• **Azure Synapse Analytics** → to process and query data efficiently
• **Power BI** → to create interactive dashboards and visual reports

Key Features of the System
• Cloud-based storage that supports scalability and flexibility
• Efficient data processing using integrated Azure services
• User-friendly and interactive data visualization
• Intelligent recommendation system for sales channel optimization
• Modular pipeline design for easy implementation and extension

Objectives
• To analyze retail sales data from different regions and categories
• To identify trends and patterns in product performance
• To compare and evaluate online and offline sales channels
• To generate actionable insights for better business decisions

Problem Description and Objectives:

In the current retail environment, businesses generate vast amounts of sales data from multiple sources such as online platforms, physical stores, and digital transactions. However, despite having access to this data, many organizations struggle to effectively analyze and utilize it for decision-making.

Several key challenges are commonly faced:

• **Data Overload:** Retail systems produce large volumes of raw data, but without proper analysis tools, this data remains underutilized and difficult to interpret.
• **Limited Analytical Capability:** Traditional analysis methods often fail to uncover meaningful insights such as product demand patterns, regional performance, and sales trends.

• **High Cost of Advanced Solutions:** Many modern analytics platforms require expensive infrastructure and resources, making them less accessible for small-scale or academic projects.

• **Complex System Setup:** Technologies like distributed computing frameworks require advanced technical knowledge, making implementation difficult for beginners.

• **Lack of Effective Visualization:** Without interactive dashboards, it becomes challenging for decision-makers to understand insights and take appropriate actions.

Due to these limitations, organizations are unable to fully leverage their data, resulting in inefficient strategies, reduced profitability, and missed opportunities for growth.

Primary Objectives
• To analyze retail sales data across different regions and product categories
• To identify patterns and trends in sales performance
• To compare the effectiveness of online and offline sales channels
• To generate useful insights for improving business strategies

Secondary Objectives
• To use Azure Data Lake Storage for data management
• To process and analyze data using Azure Synapse Analytics
• To visualize insights through Power BI dashboards

Technical Objectives
• To perform data cleaning and transformation
• To apply analytical techniques on sales data
• To design an efficient cloud-based data pipeline
• To ensure cost-effective implementation

Outcome Objectives
• To support data-driven decision-making
• To improve sales and marketing strategies
• To gain better understanding of retail performance and trends

LITERATURE SURVEY

The rapid expansion of retail businesses and digital commerce has resulted in a significant increase in sales data. Traditional methods of data processing are no longer sufficient to handle such large and continuously growing datasets. As a result, cloud computing and data analytics have emerged as essential technologies for managing and analyzing retail data efficiently.

Retail analytics has become an important area of study as it helps organizations understand sales performance, customer demand, and market trends. By analyzing transactional data, businesses can identify high-performing products, optimize inventory, and improve overall decision-making. With the adoption of cloud platforms, organizations can process large volumes of data in a scalable and cost-effective manner.

Cloud Computing in Retail Data Management

Cloud computing has transformed the way retail data is stored and processed. It provides flexible and scalable infrastructure that eliminates the need for maintaining physical hardware. Platforms like Microsoft Azure offer services such as Azure Data Lake Storage, which allows businesses to store large datasets securely and access them efficiently.

Research studies highlight that cloud-based storage solutions improve data availability, reliability, and performance. These platforms also support integration with analytics tools, enabling faster data processing and real-time insights.

Data Analytics in Retail Systems

Data analytics plays a crucial role in extracting meaningful insights from retail datasets. Analytical techniques help in identifying sales patterns, understanding product demand, and analyzing regional performance. By using structured data analysis, businesses can make informed decisions and improve operational efficiency.

Modern analytics tools allow both batch processing and real-time analysis, making it easier to handle large-scale retail data. These tools provide better accuracy and faster computation compared to traditional methods.

Sales Pattern Analysis and Decision Support

Analyzing sales patterns is essential for understanding business performance. Studies show that sales data can reveal trends such as seasonal demand, regional preferences, and product popularity. Identifying these patterns helps businesses optimize pricing strategies and improve inventory management.

Rule-based analysis techniques are commonly used to compare different sales channels and identify the most effective one. Such methods are simple, interpretable, and useful for decision

support systems.

Data Visualization and Business Intelligence

Visualization plays a key role in transforming raw data into understandable insights. Graphical representations such as bar charts, line graphs, and pie charts help users quickly interpret complex data. Interactive dashboards allow users to explore data dynamically and gain deeper insights.

Business intelligence tools like Power BI enable the creation of interactive dashboards that support decision-making. These tools allow users to filter data, compare metrics, and identify trends easily.

Comparative Analysis of Traditional and Cloud-Based Systems

Research comparing traditional data processing systems with cloud-based solutions shows that cloud platforms offer significant advantages. Traditional systems often struggle with scalability and require high maintenance costs. In contrast, cloud-based systems provide better performance, flexibility, and cost efficiency.

Cloud platforms also allow easy integration with analytics and visualization tools, making them more suitable for modern data-driven applications.

Conclusion

The literature clearly indicates that cloud computing combined with data analytics provides an effective solution for handling large retail datasets. Cloud platforms enable efficient storage, faster processing, and scalable analytics, while visualization tools enhance understanding of data insights.

This project builds on these concepts by implementing a cloud-based retail sales analytics system using Azure services and Power BI. By analyzing sales patterns, comparing channels, and generating recommendations, the system demonstrates the practical application of cloud-based analytics in retail environments.

HARDWARE AND SOFTWARE REQUIREMENTS:

Hardware Requirements

The proposed system does not require advanced hardware and can be implemented on a standard computing device.

• **Processor:** Intel i3 / i5 / i7 or equivalent
• **RAM:** Minimum 4 GB (8 GB recommended for better performance)
• **Storage:** At least 10–20 GB of free disk space
• **Internet Connection:** Required for accessing cloud services and online tools

Software Requirements

The system is developed using a combination of cloud-based platforms and data analytics tools that are efficient, accessible, and cost-effective.

Development and Data Processing Tools
• **Google Colab** – Used for data cleaning, preprocessing, and initial analysis using Python

Cloud Platform
• **Microsoft Azure**
– Azure Data Lake Storage Gen2 – for storing processed datasets
– Azure Synapse Analytics – for data processing and querying

Visualization Tool
• **Power BI Desktop** – Used to create interactive dashboards and visual reports

Programming Language
• **Python** – Used in Google Colab for data preparation
• **DAX (Data Analysis Expressions)** – Used in Power BI for calculations and recommendations

Libraries and Frameworks
The following Python libraries are used in Google Colab:
• **Pandas** – For data cleaning and manipulation
• **NumPy** – For numerical operations
• **Matplotlib / Seaborn** – For basic data visualization during preprocessing

IMPLEMENTATION

The **Retail Sales Analytics Dashboard using Azure Cloud** is developed through a structured pipeline that includes data collection, preprocessing, transformation, cloud integration, and visualization. Each stage is designed to ensure efficient data handling and accurate insight generation.

Step 1: Data Collection

Retail sales data is collected in a structured format such as CSV or Excel files. The dataset includes attributes like region, item type, sales channel (online/offline), revenue, cost, units sold, and order date.

Purpose:

• To gather raw sales data for analysis

• To ensure data is organized and ready for processing

Step 2: Data Preprocessing

The collected dataset is imported into **Google Colab** for cleaning and preparation using Python.

Activities performed:

• Handling missing or inconsistent values

• Removing duplicate entries

• Converting data types (dates, numerical values)

• Creating derived columns such as profit or total revenue

Outcome:

• Cleaned and structured dataset suitable for further analysis

Step 3: Data Transformation

After cleaning, the dataset is transformed to make it more suitable for analytical operations.

Operations include:

• Grouping data based on regions, products, and channels

• Aggregating total revenue and units sold

• Organizing data based on time intervals (monthly/yearly trends)

Outcome:

• Data becomes structured and easier to analyze

Step 4: Data Analysis

In this phase, analytical techniques are applied to extract useful insights from the dataset.

Key analyses:

• Region-wise sales performance

• Product category analysis

• Online vs offline channel comparison

• Revenue and profit trend analysis

Outcome:

• Identification of meaningful business insights and trends

Step 5: Cloud Storage Integration

The processed dataset is uploaded to **Azure Data Lake Storage Gen2** for secure and scalable storage.

Process:

• Create a storage account in Azure

• Create a container (file system)

• Upload the processed dataset

Benefits:

• Scalable cloud storage

• Secure and reliable data access

• Cost-efficient data management

Step 6: Data Visualization

The stored dataset is connected to **Power BI** to build interactive dashboards.

Visualizations include:

• Sales trend analysis (line charts)

• Region-wise performance (bar charts)

• Product distribution (pie charts)

• Channel comparison (online vs offline)

• Recommendation tables

Outcome:

• Clear and interactive representation of sales data

Step 7: Insight Generation

The dashboards are analyzed to derive actionable insights such as:

• Identification of top-performing regions

• Best-selling product categories

• Comparison of online and offline sales performance

• Recommendation of optimal sales channels (online/offline/both)

Outcome:

• Supports effective business decision-making

Step 8: Reporting

Final insights are presented through Power BI dashboards and reports.

Purpose:

• To communicate findings clearly

• To support data-driven strategies and decisions

EXPEREMENTATION AND CODE:

Experimentation

The experimentation phase involves applying analytical techniques on retail sales data to derive meaningful insights. The system was evaluated using sample datasets containing sales information such as region, product category, sales channel, revenue, and transaction details.

Experiment Setup

• The dataset was imported and processed using **Google Colab**
• Data preprocessing steps were performed to ensure data quality
• Analytical methods were applied to identify trends and patterns in sales data

Experiments Conducted

1. Data Cleaning Experiment

• Eliminated missing and inconsistent values
• Removed duplicate records
• Standardized data formats for consistency
Result: Enhanced data accuracy and improved reliability for analysis

2. Sales Segmentation Experiment

• Products were categorized based on revenue performance
• Groups such as high-performing, average-performing, and low-performing items were identified
Result: Clear classification of product performance levels

3. Sales Pattern Analysis

• Examined frequently sold product categories
• Analyzed trends based on time and region
Result: Identification of popular products and high-demand periods

4. Revenue Analysis

• Calculated total revenue and average sales values

• Analyzed contribution of different regions and product categories

Result: Better understanding of revenue distribution across the dataset

Cloud Integration Experiment

• Processed dataset was uploaded to **Azure Data Lake Storage Gen2**

• Data accessibility and storage scalability were verified

Result: Successful cloud storage with reliable data retrieval

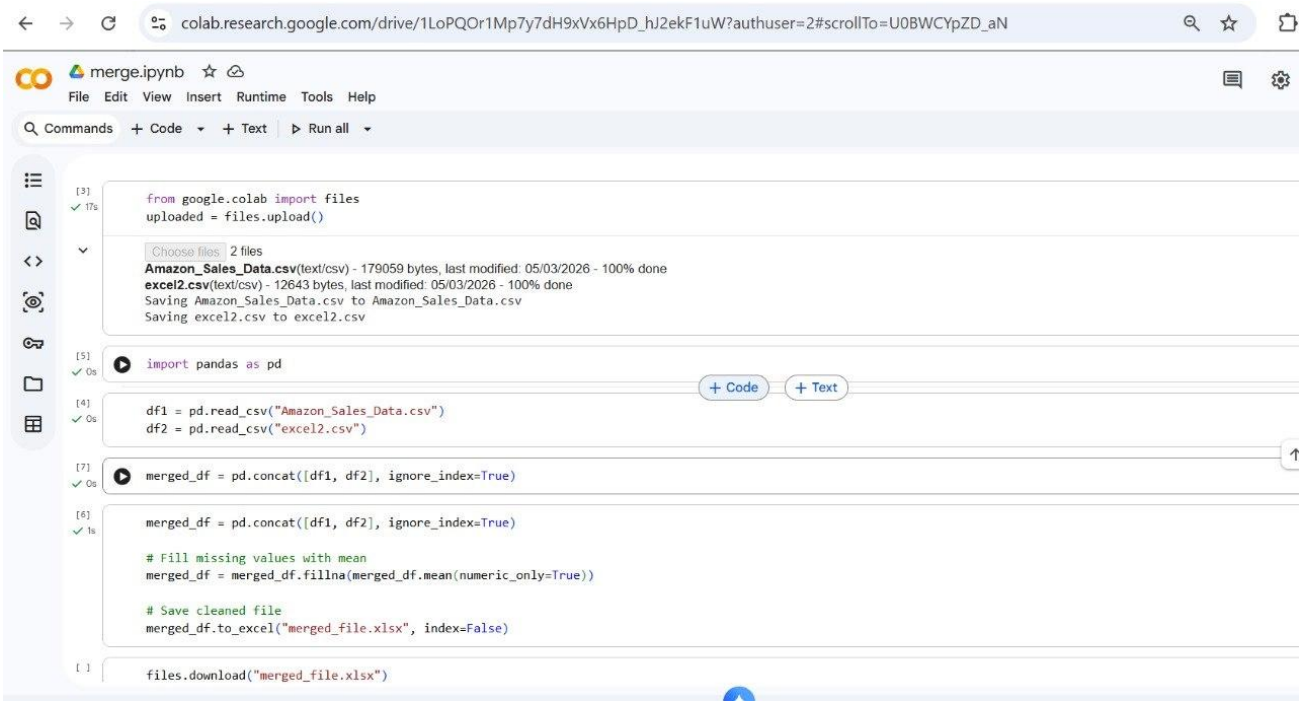
Visualization Experiment

• Dataset was connected to **Power BI**

• Interactive dashboards and reports were created

Result: Clear and effective visualization of insights for decision-making

Raw Data Processing in Google Colab



cbdas project.ipynb

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Commands + Code + Text Run all

[18]

```python
print("\nDescriptive statistics for numeric columns:")
display(merged_df.describe())
```

```
Descriptive statistics for numeric columns:
```

	Order_ID	Total_Cost	Total_Revenue	Units_Sold	Order ID	Units Sold	Unit Price	Unit Cost	Total Revenue	Total Cost	Total Profit
count	1.816000e+03	1.816000e+03	1.816000e+03	1816.000000	1.816000e+03	1816.000000	1816.000000	1816.000000	1.816000e+03	1.816000e+03	1.816000e+03
mean	5.451249e+08	9.430244e+05	1.409165e+06	5317.057895	5.550204e+08	5128.710000	276.761300	191.048000	1.373488e+06	9.318057e+05	4.416820e+05
std	2.453159e+08	1.043378e+06	1.410571e+06	2742.893784	6.086661e+07	652.650984	55.022493	43.955961	3.409892e+05	2.531534e+05	1.024204e+05
min	1.146066e+08	3.612240e+03	4.870260e+03	124.000000	1.146066e+08	124.000000	9.330000	6.920000	4.870260e+03	3.612240e+03	1.258020e+03
25%	3.457186e+08	1.970483e+05	3.870022e+05	3482.000000	5.550204e+08	5128.710000	276.761300	191.048000	1.373488e+06	9.318057e+05	4.416820e+05
50%	5.451249e+08	4.489777e+05	8.023338e+05	5430.000000	5.550204e+08	5128.710000	276.761300	191.048000	1.373488e+06	9.318057e+05	4.416820e+05
75%	7.288153e+08	1.373244e+06	2.011150e+06	7450.000000	5.550204e+08	5128.710000	276.761300	191.048000	1.373488e+06	9.318057e+05	4.416820e+05
max	9.940222e+08	4.509794e+06	5.997055e+06	9925.000000	9.940222e+08	9925.000000	668.270000	524.960000	5.997055e+06	4.509794e+06	1.719922e+06

[19]

```python
print("\nNumber of unique values per column:")
display(merged_df.nunique())
```

```
Number of unique values per column:
```

	0
Country	78
Item_Type	12
Order_Date	100
Order_ID	101
Order_Priority	4
Region	8
Sales_Channel	2

Variables Terminal 12:10 PM

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[20]

```python
# List of column pairs to consolidate (target_col, source_col)
column_pairs_to_consolidate = [
    ('Item_Type', 'Item Type'),
    ('Order_Date', 'Order Date'),
    ('Order_ID', 'Order ID'),
    ('Order_Priority', 'Order Priority'),
    ('Sales_Channel', 'Sales Channel'),
    ('Ship_Date', 'Ship Date'),
    ('Units_Sold', 'Units Sold'),
    ('Total_Cost', 'Total Cost'),
    ('Total_Revenue', 'Total Revenue')
]

for target_col, source_col in column_pairs_to_consolidate:
    if source_col in merged_df.columns:
        # Fill NaNs in the target column with values from the source column
        # Use .astype(str) for object columns to ensure consistency before fillna
        # For numeric columns, fillna directly
        if merged_df[target_col].dtype == 'object' or merged_df[source_col].dtype == 'object':
            merged_df[target_col] = merged_df[target_col].astype(str).replace('nan', pd.NA).fillna(merged_df[source_col].astype(str).replace('nan', pd.NA))
        else:
            merged_df[target_col] = merged_df[target_col].fillna(merged_df[source_col])

        # Drop the source column after consolidation
        merged_df = merged_df.drop(columns=[source_col])

# Display info again to confirm consolidation and check for remaining NaNs
print("Information about merged_df after consolidating columns:")
merged_df.info()

print("\nNumber of unique values per column after consolidating columns:")
display(merged_df.nunique())
```

```
Information about merged_df after consolidating columns:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1816 entries, 0 to 1815
Data columns (total 14 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Country         1810 non-null   object
 1   Item_Type       1756 non-null   object
 2   Order_Date      1810 non-null   object
 3   Order ID        1816 non-null   float64
```

Variables Terminal 12:10 PM

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```
 1   Item_Type       1756 non-null   object
 2   Order_Date      1810 non-null   object
 3   Order_ID        1816 non-null   float64
 4   Order_Priority  1810 non-null   object
 5   Region          1810 non-null   object
 6   Sales_Channel   1810 non-null   object
 7   Ship_Date       1810 non-null   object
 8   Total_Cost      1816 non-null   float64
 9   Total_Revenue   1816 non-null   float64
 10  Units_Sold      1816 non-null   float64
 11  Unit Price      1816 non-null   float64
 12  Unit Cost       1816 non-null   float64
 13  Total Profit    1816 non-null   float64
dtypes: float64(7), object(7)
memory usage: 198.8+ KB

Number of unique values per column after consolidating columns:
```

	0
Country	78
Item_Type	12
Order_Date	200
Order_ID	101
Order_Priority	4
Region	8
Sales_Channel	2
Ship_Date	198
Total_Cost	101
Total_Revenue	101
Units_Sold	100
Unit Price	13
Unit Cost	13
Total Profit	101

dtype: int64

Variables Terminal 12:10 PM

cbdas project.ipynb

File Edit View Insert Runtime Tools Help

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[26]

```python
import matplotlib.pyplot as plt
import seaborn as sns

# Set the aesthetic style of the plots
sns.set_style("whitegrid")

# Create subplots for each histogram
fig, axes = plt.subplots(1, 3, figsize=(20, 6))

# Histogram for Total_Revenue
sns.histplot(merged_df['Total_Revenue'], kde=True, ax=axes[0], color='skyblue')
axes[0].set_title('Distribution of Total Revenue')
axes[0].set_xlabel('Total Revenue')
axes[0].set_ylabel('Frequency')

# Histogram for Total_Cost
sns.histplot(merged_df['Total_Cost'], kde=True, ax=axes[1], color='lightcoral')
axes[1].set_title('Distribution of Total Cost')
axes[1].set_xlabel('Total Cost')
axes[1].set_ylabel('Frequency')

# Histogram for Total_Profit
sns.histplot(merged_df['Total Profit'], kde=True, ax=axes[2], color='lightgreen')
axes[2].set_title('Distribution of Total Profit')
axes[2].set_xlabel('Total Profit')
axes[2].set_ylabel('Frequency')

plt.tight_layout()
plt.show()
```

cbdas project.ipynb

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[26]

```python
import matplotlib.pyplot as plt
import seaborn as sns

# Set the aesthetic style of the plots
sns.set_style("whitegrid")

# Create subplots for each histogram
fig, axes = plt.subplots(1, 3, figsize=(20, 6))

# Histogram for Total_Revenue
sns.histplot(merged_df['Total_Revenue'], kde=True, ax=axes[0], color='skyblue')
axes[0].set_title('Distribution of Total Revenue')
axes[0].set_xlabel('Total Revenue')
axes[0].set_ylabel('Frequency')

# Histogram for Total_Cost
sns.histplot(merged_df['Total_Cost'], kde=True, ax=axes[1], color='lightcoral')
axes[1].set_title('Distribution of Total Cost')
axes[1].set_xlabel('Total Cost')
axes[1].set_ylabel('Frequency')

# Histogram for Total_Profit
sns.histplot(merged_df['Total Profit'], kde=True, ax=axes[2], color='lightgreen')
axes[2].set_title('Distribution of Total Profit')
axes[2].set_xlabel('Total Profit')
axes[2].set_ylabel('Frequency')

plt.tight_layout()
plt.show()
```

cbdas project.ipynb

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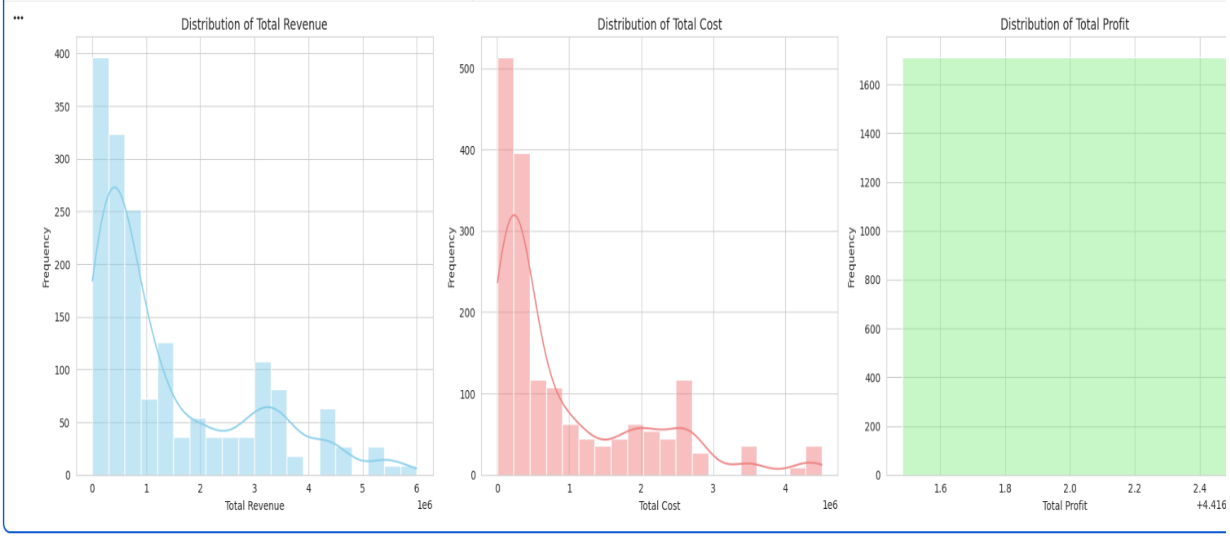
[26]

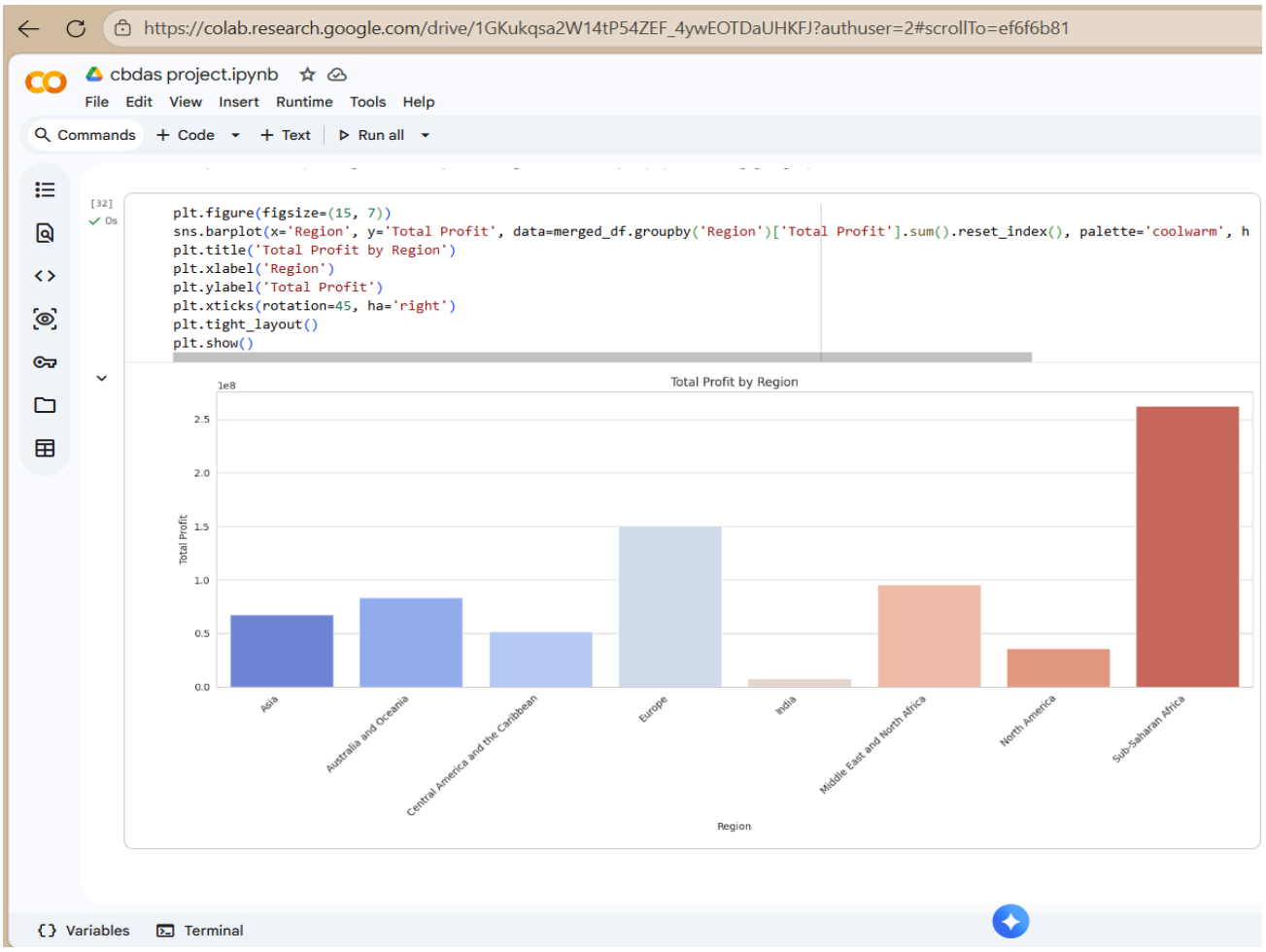
```python
plt.tight_layout()
plt.show()
```



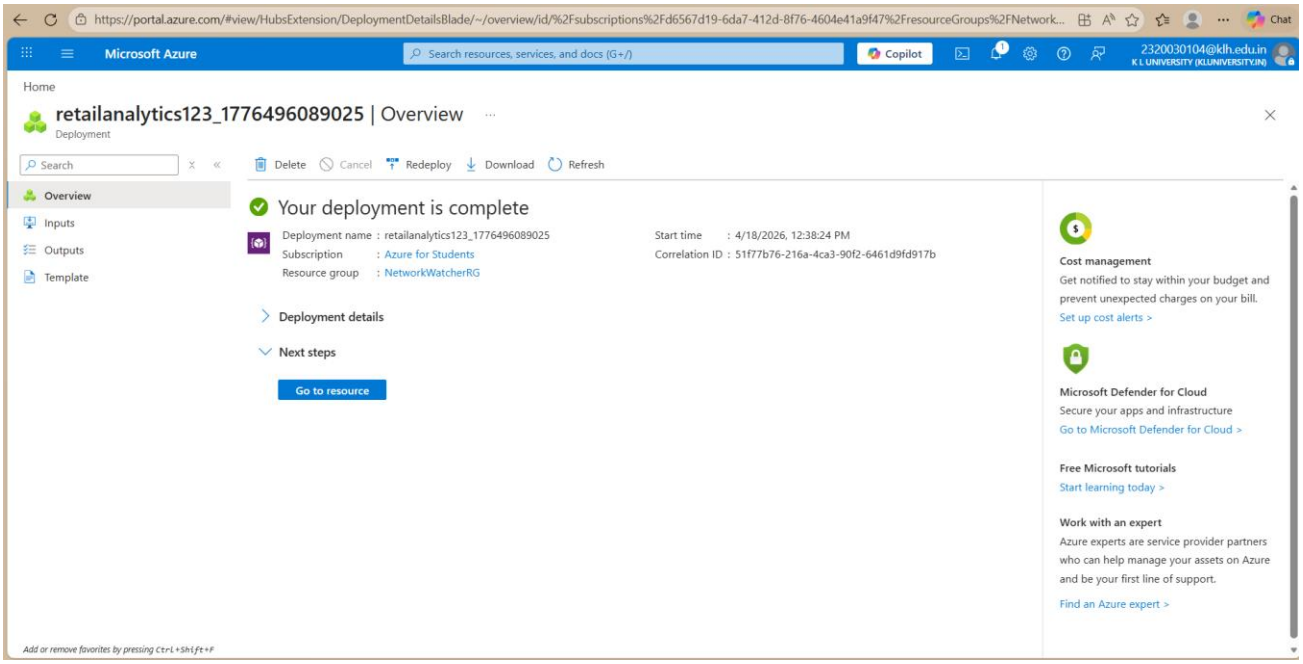
[30]

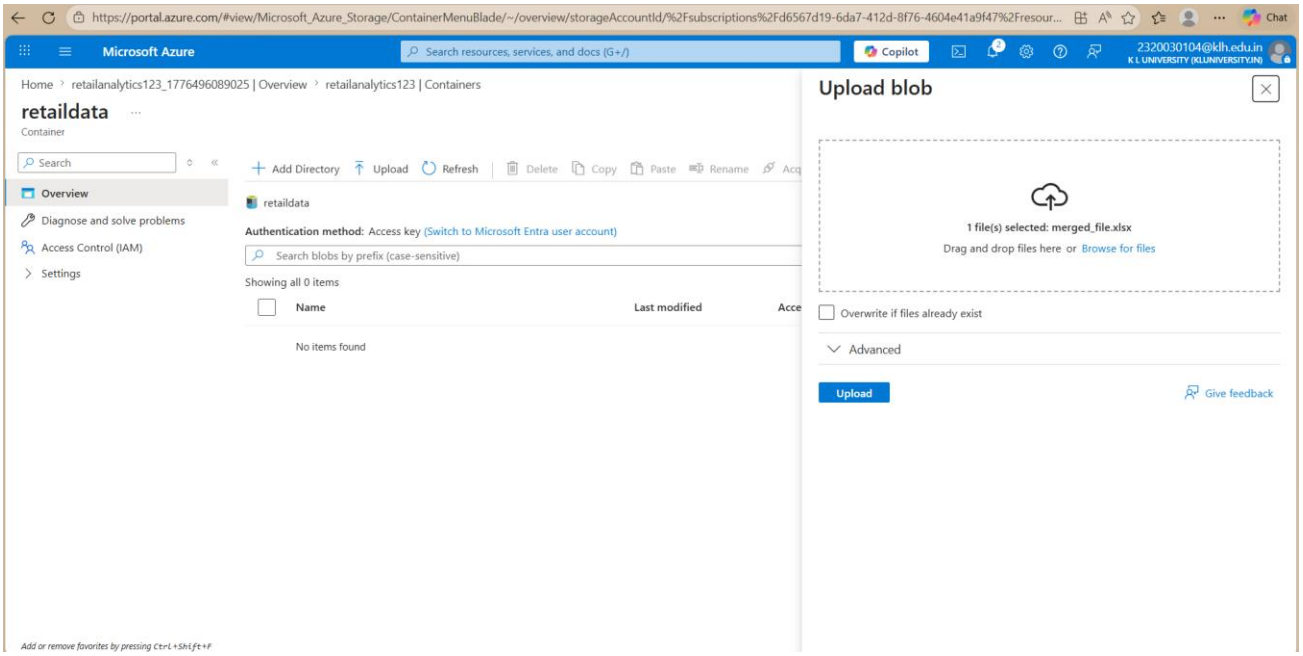
```python
import matplotlib.pyplot as plt
import seaborn as sns


# Histogram for Total_Cost
sns.histplot(merged_df['Total_Cost'], kde=True, ax=axes[1], color='lightcoral')
axes[1].set_title('Distribution of Total Cost')
```

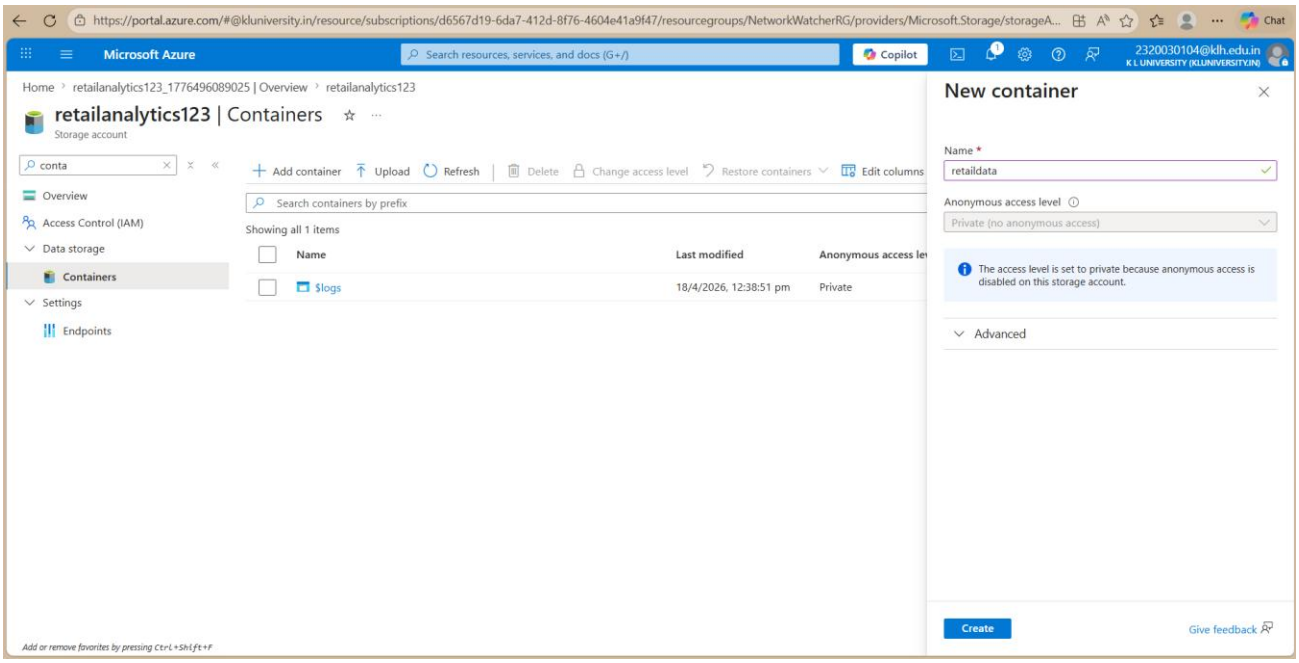
Variables Terminal 12:15 PM

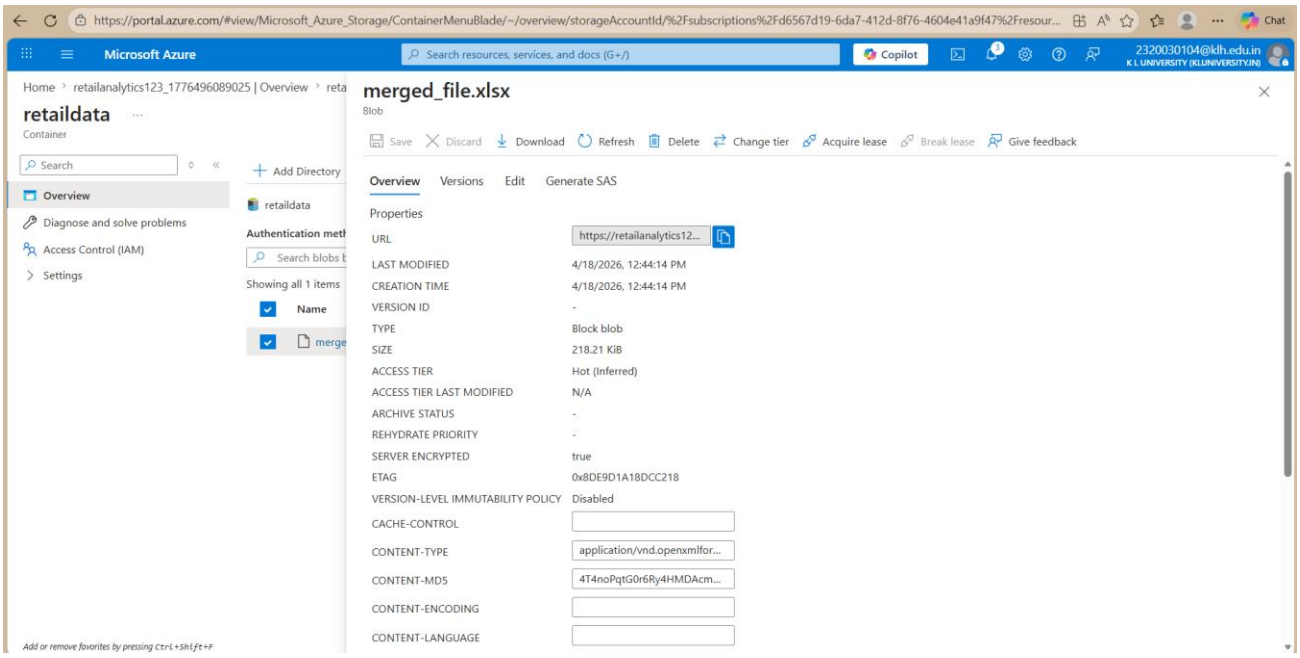

https://colab.research.google.com/drive/1GKukqsa2W14tP54ZEF_4ywEOTDaUHKFJ?authuser=2#scrollTo=ef6f6b81
cbdas project.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
[32]
plt.figure(figsize=(15, 7))
sns.barplot(x='Region', y='Total Profit', data=merged_df.groupby('Region')['Total Profit'].sum().reset_index(), palette='coolwarm', h
plt.title('Total Profit by Region')
plt.xlabel('Region')
plt.ylabel('Total Profit')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
Total Profit by Region
1e8
2.5
2.0
1.5
1.0
0.5
0.0
Total Profit
Asia
Australia and Oceania
Central America and the Caribbean
Europe
India
Middle East and North Africa
North America
Sub-Saharan Africa
Region
Variables
Terminal

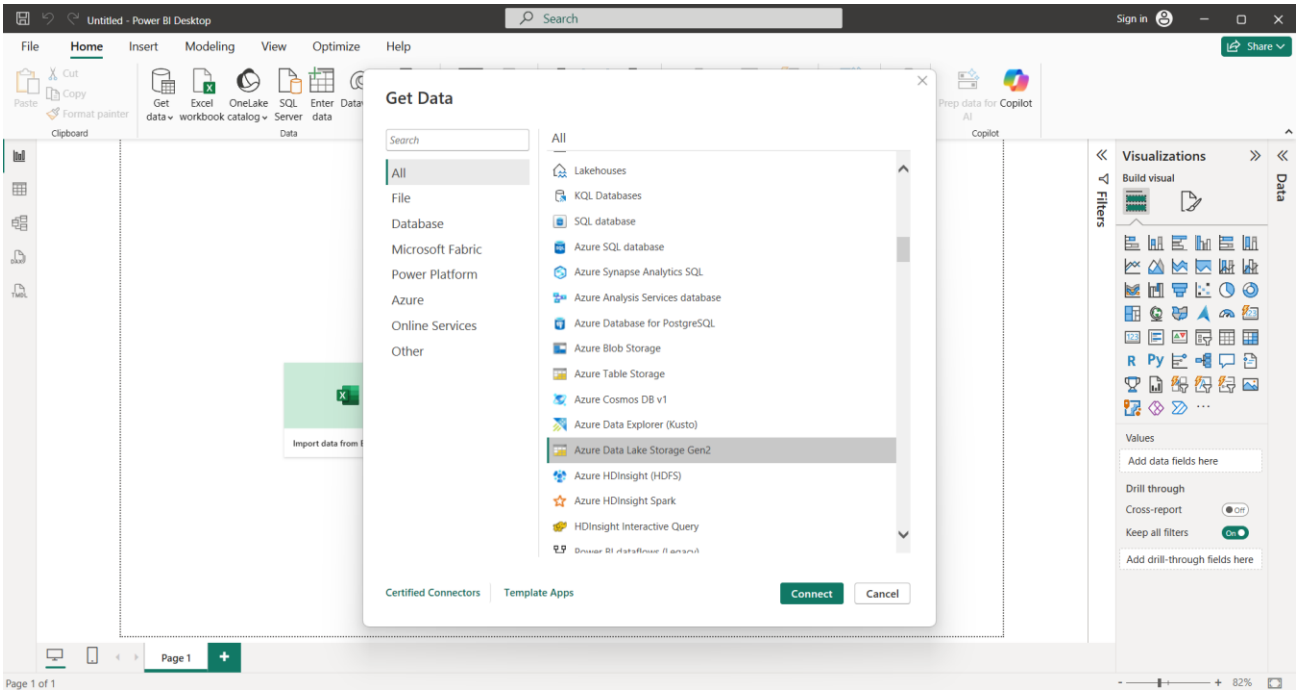
Data Storage in Azure Portal

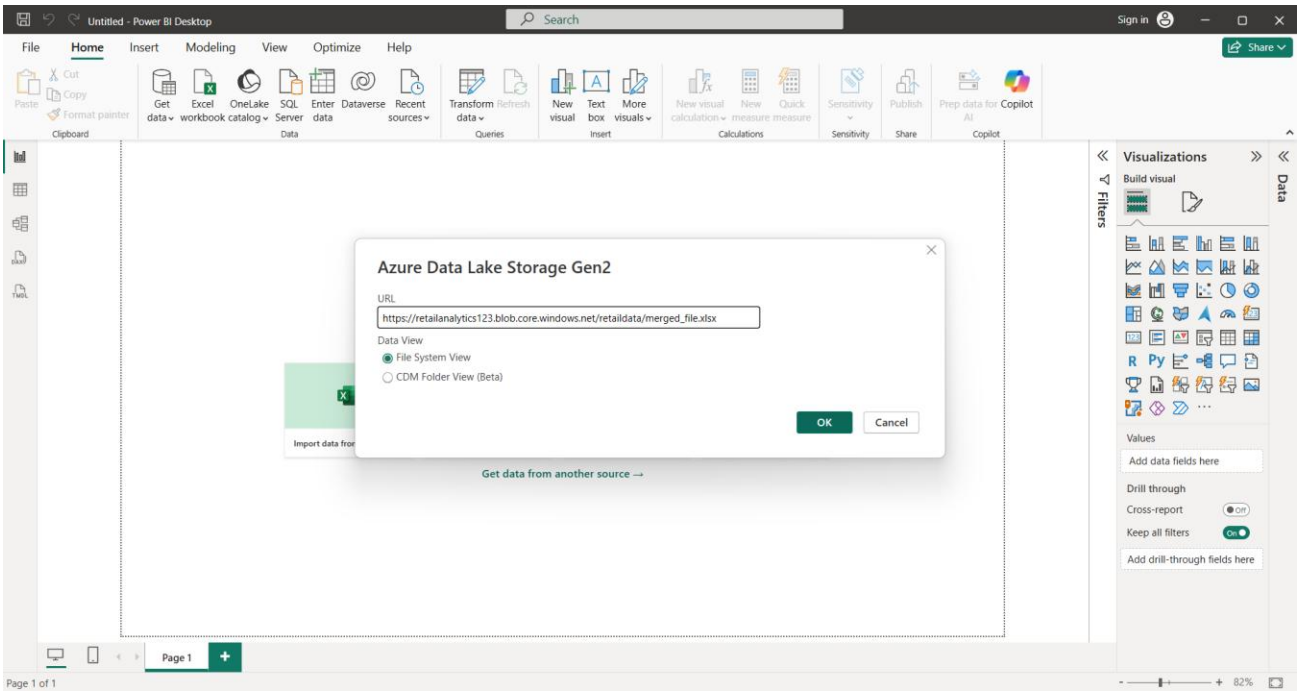


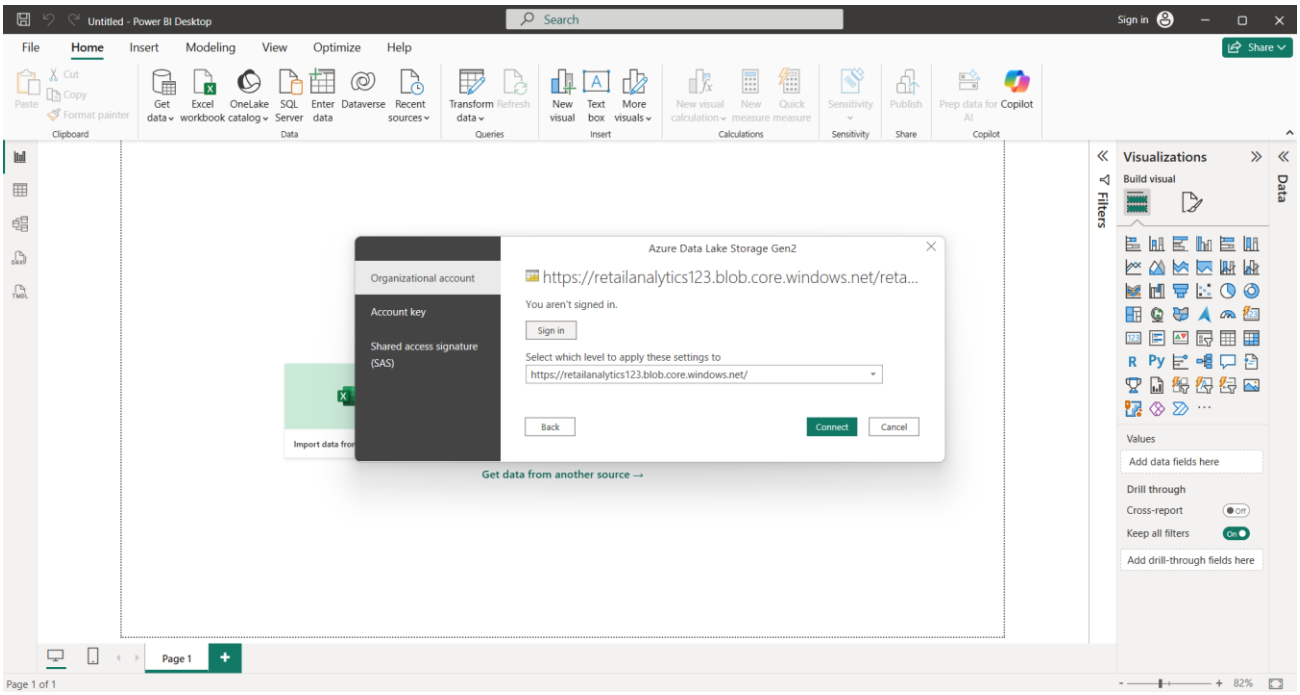



Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
2320030104@klh.edu.in
K L UNIVERSITY (KLUNIVERSITY.IN)
Home > retailanalytics123_1776496089025 | Overview > retailanalytics123
retailanalytics123 | Containers
Storage account
conta
Overview
Access Control (IAM)
Data storage
Containers
Settings
Endpoints
Add container
Upload
Refresh
Delete
Change access level
Restore containers
Edit columns
Search containers by prefix
Showing all 1 items
Name
Last modified
Anonymous access lev
$logs
18/4/2026, 12:38:51 pm
Private
New container
Name *
retaildata
Anonymous access level
Private (no anonymous access)
The access level is set to private because anonymous access is disabled on this storage account.
Advanced
Create
Give feedback
Add or remove favorites by pressing Ctrl+Shift+F


Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
2320030104@klh.edu.in
K L UNIVERSITY (KLUNIVERSITY.IN)
Home > retailanalytics123_1776496089025 | Overview > reta
retaildata
Container
Search
Overview
Diagnose and solve problems
Access Control (IAM)
Settings
Add Directory
retaildata
Authentication meth
Search blobs b
Showing all 1 items
Name
merge
merged_file.xlsx
Blob
Save
Discard
Download
Refresh
Delete
Change tier
Acquire lease
Break lease
Give feedback
Overview
Versions
Edit
Generate SAS
Properties
URL
https://retailanalytics12...
LAST MODIFIED
4/18/2026, 12:44:14 PM
CREATION TIME
4/18/2026, 12:44:14 PM
VERSION ID
-
TYPE
Block blob
SIZE
218.21 KiB
ACCESS TIER
Hot (Inferred)
ACCESS TIER LAST MODIFIED
N/A
ARCHIVE STATUS
-
REHYDRATE PRIORITY
-
SERVER ENCRYPTED
true
ETAG
0x8DE9D1A18DCC218
VERSION-LEVEL IMMUTABILITY POLICY
Disabled
CACHE-CONTROL
CONTENT-TYPE
application/vnd.openxmlfor...
CONTENT-MD5
4T4noPqtG0r6Ry4HMDAcm...
CONTENT-ENCODING
CONTENT-LANGUAGE
Add or remove favorites by pressing Ctrl+Shift+F


Untitled - Power BI Desktop
Search
Sign in
File
Home
Insert
Modeling
View
Optimize
Help
Share
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Copilot
Get Data
Search
All
File
Database
Microsoft Fabric
Power Platform
Azure
Online Services
Other
Lakehouses
KQL Databases
SQL database
Azure SQL database
Azure Synapse Analytics SQL
Azure Analysis Services database
Azure Database for PostgreSQL
Azure Blob Storage
Azure Table Storage
Azure Cosmos DB v1
Azure Data Explorer (Kusto)
Azure Data Lake Storage Gen2
Azure HDInsight (HDFS)
Azure HDInsight Spark
HDInsight Interactive Query
Certified Connectors
Template Apps
Connect
Cancel
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Page 1
Page 1 of 1
82%


Untitled - Power BI Desktop
Search
Sign in
File
Home
Insert
Modeling
View
Optimize
Help
Share
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Transform data
Refresh
New visual
Text box
More visuals
Copilot
Azure Data Lake Storage Gen2
URL
https://retailanalytics123.blob.core.windows.net/retaildata/merged_file.xlsx
Data View
File System View
CDM Folder View (Beta)
OK
Cancel
Get data from another source →
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Page 1
Page 1 of 1
82%


Untitled - Power BI Desktop
Search
Sign in
File
Home
Insert
Modeling
View
Optimize
Help
Share
Paste
Cut
Copy
Format painter
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Publish
Share
Prep data for AI
Copilot
Azure Data Lake Storage Gen2
Organizational account
Account key
Shared access signature (SAS)
https://retailanalytics123.blob.core.windows.net/reta...
You aren't signed in.
Sign in
Select which level to apply these settings to
https://retailanalytics123.blob.core.windows.net/
Back
Connect
Cancel
Get data from another source →
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Filters
Data
Page 1
Page 1 of 1
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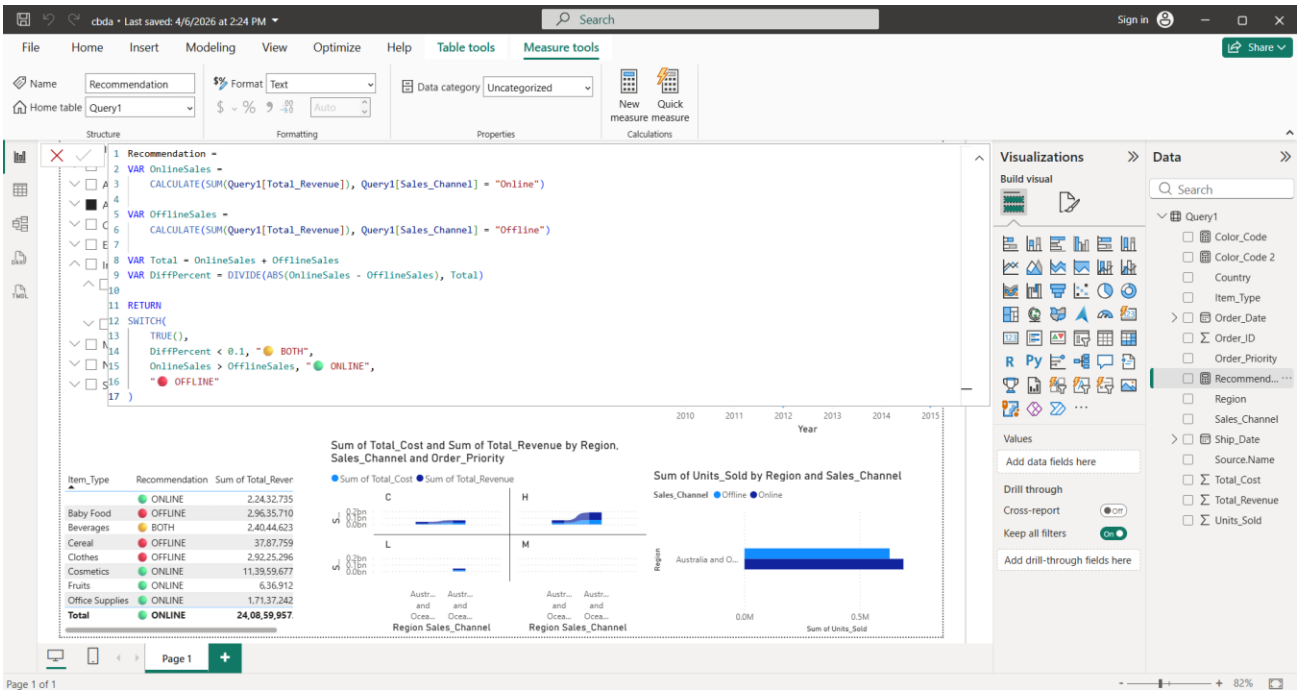
cbda • Last saved: 4/6/2026 at 2:24 PM
Search
Sign in
File
Home
Insert
Modeling
View
Optimize
Help
Table tools
Measure tools
Share
Name
Recommendation
Home table
Query1
Structure
Format
Text
Formatting
Data category
Uncategorized
Properties
New measure
Quick measure
Calculations
1 Recommendation =
2 VAR OnlineSales =
3 CALCULATE(SUM(Query1[Total_Revenue]), Query1[Sales_Channel] = "Online")
4
5 VAR OfflineSales =
6 CALCULATE(SUM(Query1[Total_Revenue]), Query1[Sales_Channel] = "Offline")
7
8 VAR Total = OnlineSales + OfflineSales
9 VAR DiffPercent = DIVIDE(ABS(OnlineSales - OfflineSales), Total)
10
11 RETURN
12 SWITCH(
13 TRUE(),
14 DiffPercent < 0.1, " BOTH",
15 OnlineSales > OfflineSales, " ONLINE",
16 " OFFLINE"
17)
Item_Type
Recommendation
Sum of Total_Rever
Baby Food
Beverages
Cereal
Clothes
Cosmetics
Fruits
Office Supplies
Total
ONLINE
OFFLINE
BOTH
24,08,59,957
Sum of Total_Cost and Sum of Total_Revenue by Region, Sales_Channel and Order_Priority
Sum of Total_Cost
Sum of Total_Revenue
Region Sales_Channel
Year
Sum of Units_Sold by Region and Sales_Channel
Sales_Channel
Offline
Online
Region
Sum of Units_Sold
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Data
Search
Query1
Color_Code
Color_Code 2
Country
Item_Type
Order_Date
Order_ID
Order_Priority
Recommend...
Region
Sales_Channel
Ship_Date
Source.Name
Total_Cost
Total_Revenue
Units_Sold
Page 1
Page 1 of 1
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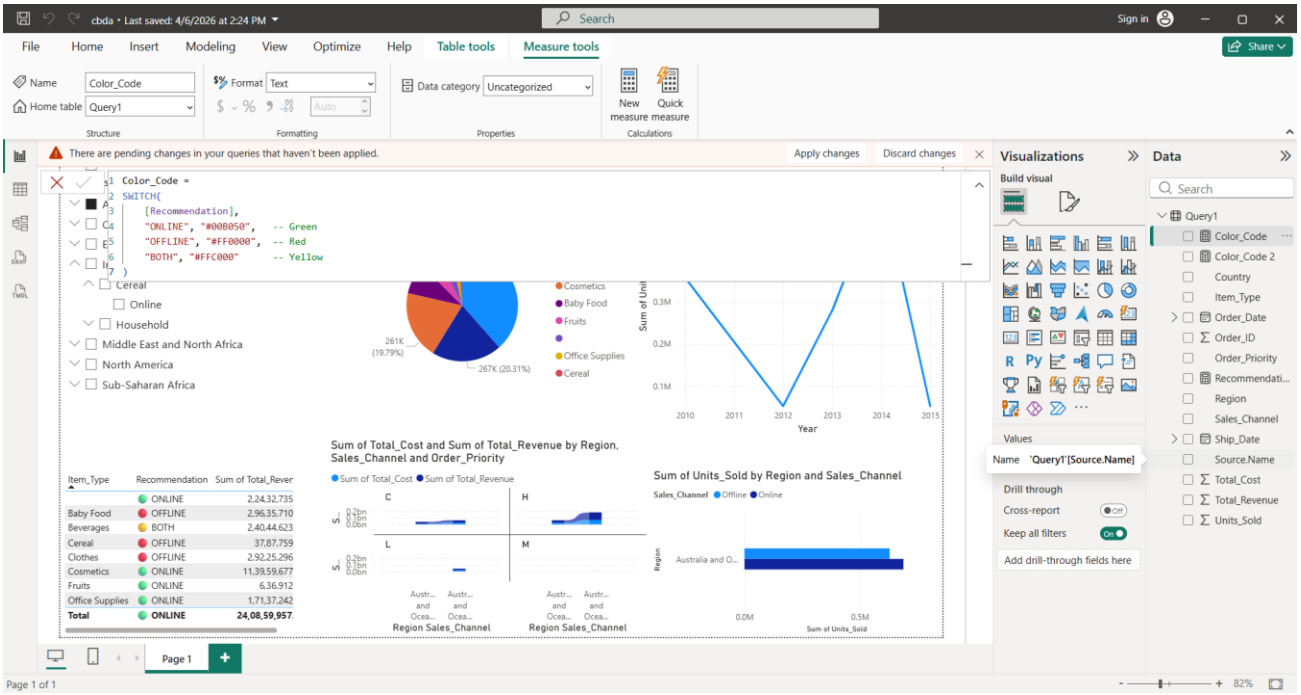
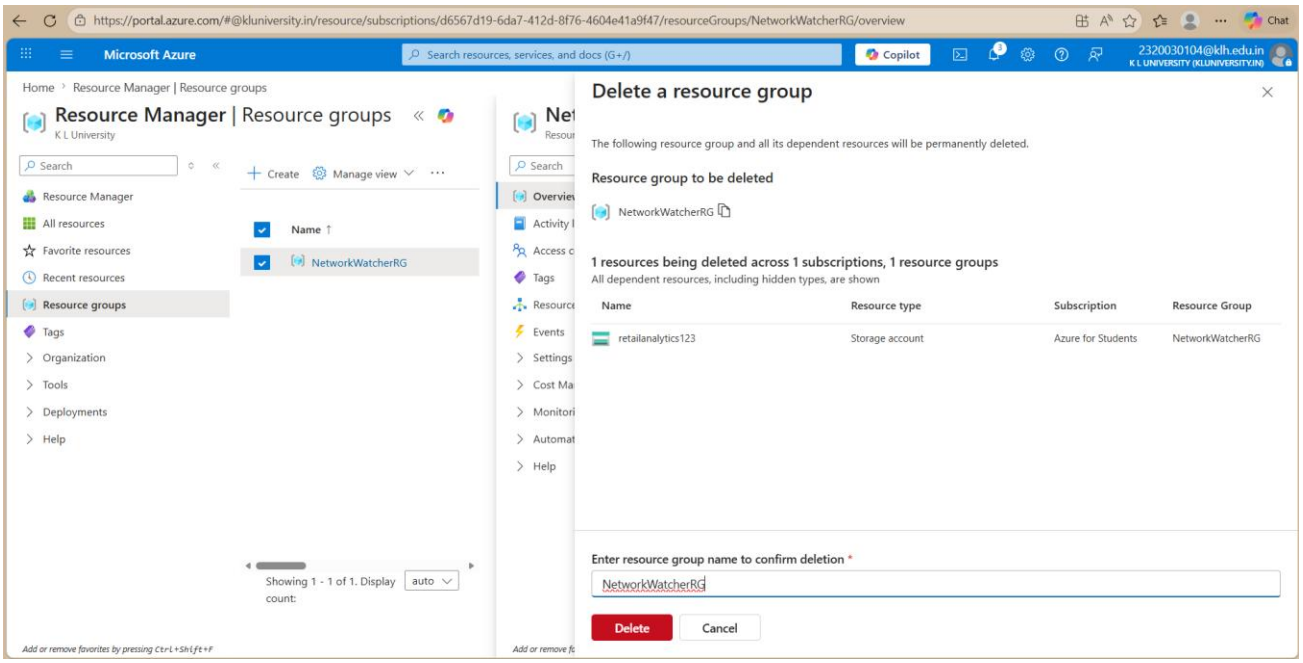


Item_Type	Recommendation	Sum of Total_Rever
	ONLINE	2,24,32,735
Baby Food	OFFLINE	2,96,35,710
Beverages	BOTH	2,40,44,623
Cereal	OFFLINE	37,87,759
Clothes	OFFLINE	2,92,25,296
Cosmetics	ONLINE	11,39,59,677
Fruits	ONLINE	6,36,912
Office Supplies	ONLINE	1,71,37,242
Total	**ONLINE**	**24,08,59,957**

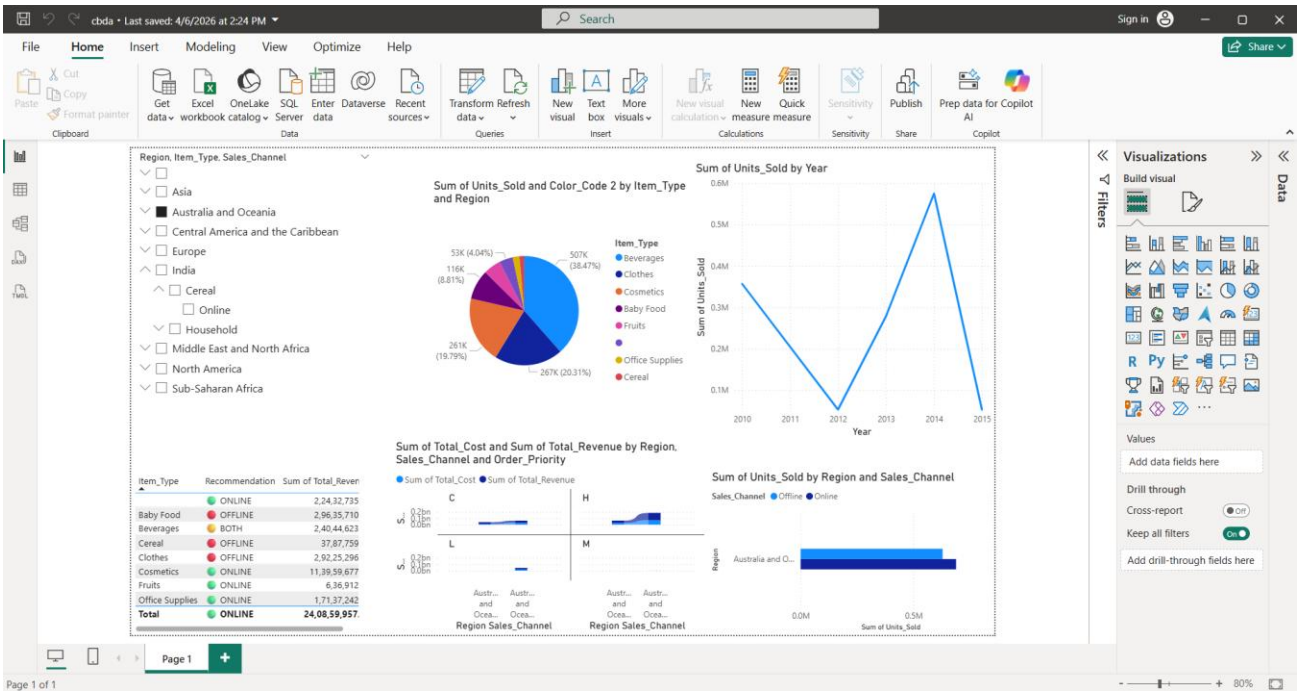


RESULT

The dataset stored in **Azure Data Lake Storage Gen2** was successfully connected to Power BI using the Azure Data Lake connector. This integration enabled seamless access to cloud-stored data for visualization and analysis without the need for intermediate steps.

Result:

- Established a direct and secure connection between Azure cloud storage and Power BI
- Enabled smooth and efficient data import with minimal latency
- Eliminated the need for manual file downloads or data transfers
- Allowed real-time or near real-time access to updated datasets
- Improved overall workflow efficiency by integrating storage and visualization platforms



CONCLUTION

The **Retail Sales Analytics Dashboard using Azure Cloud** was successfully designed and implemented to analyze retail sales data and generate meaningful business insights. The system effectively combines data preprocessing, cloud storage, and visualization into a streamlined and cost-efficient workflow.

Using **Google Colab**, the raw dataset was cleaned, transformed, and prepared for analysis in an efficient manner. Various analytical techniques were applied to identify sales trends, regional performance, and product-level insights. The processed data was then securely stored in **Azure Data Lake Storage Gen2**, ensuring scalability, reliability, and easy access.

The integration with **Power BI** enabled the development of interactive dashboards that clearly present sales patterns, channel comparisons, and performance metrics. Additionally, a DAX-based recommendation system was implemented to suggest whether products should be sold online, offline, or both, enhancing the decision-making capability of the system.

Overall, the project demonstrates a practical, scalable, and cost-effective approach to retail data analytics. It successfully transforms raw sales data into actionable insights without requiring complex infrastructure, making it suitable for academic use as well as small and medium-scale business applications.

REFRENCES

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Available at: https://learn.microsoft.com/en-us/power-bi/
This source was used to understand dashboard development, data visualization techniques, and report creation.

2. Microsoft Azure Data Lake Documentation
Microsoft Azure. *Azure Data Lake Storage Documentation.*
Available at: https://learn.microsoft.com/en-us/azure/storage/blobs/
This reference helped in understanding cloud-based storage, data management, and integration with analytics tools.

3. Python Official Documentation
Python Software Foundation. *Python Documentation.*
Available at: https://docs.python.org/3/
This documentation was referred to for Python programming concepts, data handling, and analytical operations.

4. Google Colaboratory Documentation
Google. *Google Colaboratory Documentation.*
Available at: https://colab.research.google.com/
This resource provided guidance on cloud-based Python execution, data preprocessing, and analysis workflows.

5. Azure Synapse Analytics Documentation
Microsoft Azure. *Azure Synapse Analytics Overview.*
Available at: https://learn.microsoft.com/en-us/azure/synapse-analytics/
This source was used to understand data processing, querying, and integration within the Azure ecosystem.

6. Big Data and Retail Analytics Concepts
Chen, M., Mao, S., & Liu, Y. (2014). *Big Data: A Survey.*
Available at: https://ieeexplore.ieee.org/document/6844456
This paper provided insights into large-scale data analytics and its relevance in business and retail applications.

7. Data Visualization Techniques
Few, S. (2009). *Now You See It: Simple Visualization Techniques for Quantitative Analysis.*
Available at: https://www.analyticsvidhya.com/blog/2018/01/data-visualization-best-practices/
This reference helped in understanding effective visualization methods and dashboard design principles.